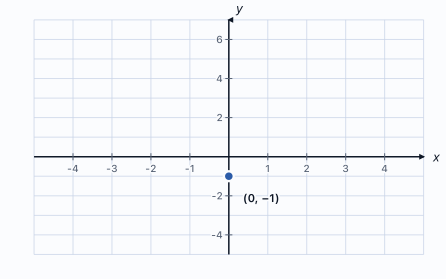
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Functions, Transformations & Arithmetic Series Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 10x + 38$ in the domain of Functions, Transformations & Arithmetic Series. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(7)$.



A 35
B 17
C 30
D 28

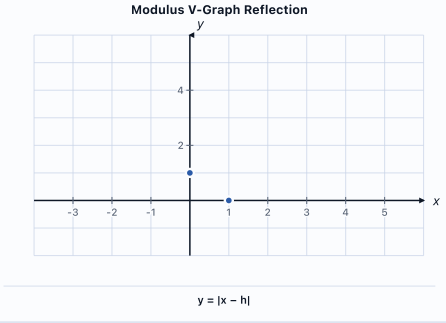
2 MCQ

A mathematical model in Functions, Transformations & Arithmetic Series is defined by the cubic function $y_2(x) = 4x^3 - 5x^2 + 15x - 10$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123
B 78
C 41
D 93

3 MCQ

In Functions, Transformations & Arithmetic Series, two displacement vectors are given by $\mathbf{u}_3 = \langle 5, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 78
B -11
C 89
D 190

4 MCQ

The linear growth profile shown in the diagram for Functions, Transformations & Arithmetic Series follows an arithmetic progression with first term $a_1 = 6$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 90

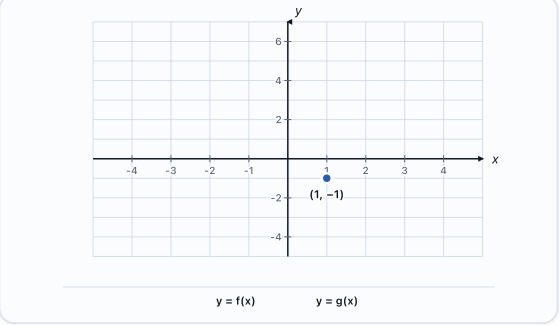
B 720

C 810

D 1350

5 MCQ

In Functions, Transformations & Arithmetic Series, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 20x)\,\mathrm{d}x$.



A 112

B 108

C 88

D 45

6 MCQ

In Functions, Transformations & Arithmetic Series, consider the quadratic function $f_6(x) = x^2 - 12x + 121$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(13)$.

A 225

B 134

C 212

D 219

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 6x^2 + 33x - 17$ studied in Functions, Transformations & Arithmetic Series. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 705

B 801

C 672

D 117

8 MCQ

In Functions, Transformations & Arithmetic Series, two displacement vectors are given by $\mathbf{u}_{8} = \langle 6, -7 \rangle$ and $\mathbf{v}_{8} = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_{8} \cdot \mathbf{v}_{8} + \|\mathbf{u}_{8}\|^{2}$.

A 13

B 98

C 85

D 168

9 MCQ

An arithmetic series has first term $a = 4$ and common difference $d = 3$. Calculate the sum S_{12} of the first 12 terms.

A 282

B 444

C 246

D 37

10 MCQ

An arithmetic series has first term $a = 5$ and common difference $d = 4$. Calculate the sum S_{12} of the first 12 terms.

A 372

B 588

C 49

D 324

11 MCQ

An arithmetic series has first term $a = 6$ and common difference $d = 5$. Calculate the sum S_{12} of the first 12 terms.

A 732

B 462

C 61

D 402

12 MCQ

An arithmetic series has first term $a = 7$ and common difference $d = 6$. Calculate the sum S_{12} of the first 12 terms.

A 480

B 73

C 552

D 876

13 MCQ

An arithmetic series has first term $a = 8$ and common difference $d = 7$. Calculate the sum S_{13} of the first 13 terms.

A 650

B 1196

C 741

D 92

14 MCQ

An arithmetic series has first term $a = 9$ and common difference $d = 8$. Calculate the sum S_{13} of the first 13 terms.

A 105

B 1365

C 741

D 845

15 MCQ

An arithmetic series has first term $a = 10$ and common difference $d = 3$. Calculate the sum S_{13} of the first 13 terms.

A 403

B 364

C 46

D 598

16 MCQ

An arithmetic series has first term $a = 11$ and common difference $d = 4$. Calculate the sum S_{13} of the first 13 terms.

A 767

B 455

C 507

D 59

17 MCQ

An arithmetic series has first term $a = 12$ and common difference $d = 5$. Calculate the sum S_{14} of the first 14 terms.

A 1078

B 623

C 77

D 693

18 MCQ

An arithmetic series has first term $a = 13$ and common difference $d = 6$. Calculate the sum S_{14} of the first 14 terms.

A 728

B 1274

C 812

D 91

19 MCQ

An arithmetic series has first term $a = 14$ and common difference $d = 7$. Calculate the sum S_{14} of the first 14 terms.

A 1470

B 833

C 931

D 105

20 MCQ

An arithmetic series has first term $a = 4$ and common difference $d = 8$. Calculate the sum S_{14} of the first 14 terms.

A 108

B 784

C 896

D 1512

21 MCQ

An arithmetic series has first term $a = 5$ and common difference $d = 3$. Calculate the sum S_{15} of the first 15 terms.

A 705

B 47

C 435

D 390

22 MCQ

An arithmetic series has first term $a = 6$ and common difference $d = 4$. Calculate the sum S_{15} of the first 15 terms.

A 510

B 930

C 62

D 570

23 MCQ

An arithmetic series has first term $a = 7$ and common difference $d = 5$. Calculate the sum S_{15} of the first 15 terms.

A 1155

B 77

C 630

D 705

24 MCQ

An arithmetic series has first term $a = 8$ and common difference $d = 6$. Calculate the sum S_{15} of the first 15 terms.

A 92

B 1380

C 750

D 840

25 MCQ

An arithmetic series has first term $a = 9$ and common difference $d = 7$. Calculate the sum S_{16} of the first 16 terms.

A 984

B 1824

C 1096

D 114